If you first look at the control group histogram in figure 7.2, it appears to be a normal-like distribution centered around –3.0 mmHg (a unit of pressure), which is a fairly insignificant reduction in blood pressure, as you would expect from a placebo. Our algorithm would be correct to predict that for any patient given a placebo, their expected blood pressure change would be –3.0 mmHg on average, even though individual patients had greater or lesser changes than that average value.

Now look at the treatment group histogram. The distribution of blood pressure change is bimodal, meaning there are two peaks, as if we had combined two separate normal distributions. The right-most mode is centered at –2.5 mmHg, much like the control group, suggesting that this subgroup within the treatment group did not benefit from the drug compared to the placebo. However, the left-most mode is centered at –22.3 mmHg, which is a very significant reduction in blood pressure. In fact, it's greater than any currently existing anti-hypertensive drug. This again indicates that there is a subgroup within the treatment group, but this subgroup strongly benefits from the drug.

If you're a physician, and a patient with hypertension walks into your office, all else being equal, should you prescribe them this new drug? If you take the expected value (the average) of the treatment group distribution, you'd only get about –13 mmHg change in blood pressure, which is between the two modes in the distribution. This is still significant compared to the placebo, but it's worse than many existing anti-hypertensives on the market. By that standard, the new drug does not appear to be very effective, despite the fact that a decent number of patients get tremendous benefit from it. Moreover, the expected value of –13 mmHg is very poorly representative of the distribution, since very few patients actually had that level of blood pressure reduction. Patients either had almost no response to the drug or a very robust response; there were very few moderate responders.

Figure 7.3 illustrates the limitations of expected values compared to seeing the full distribution. If you use the expected values of blood pressure changes for each drug, and just pick the drug with the lowest expected value in terms of blood pressure change (ignoring patient-specific